Oxford College of Emory University

Chapter 14, Part 1

Money, Liquidity, and Banking

Guilherme Paiva Pinto

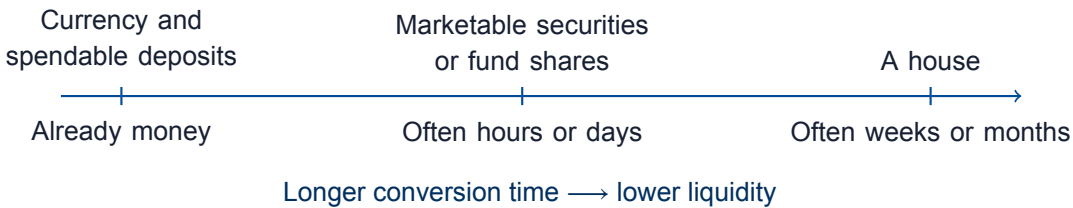
October 27, 2026

Roadmap

1. Money, liquidity, and the money supply
2. Banks, reserves, and T-accounts
3. Bank runs and safeguards
4. Moral hazard and shadow banking

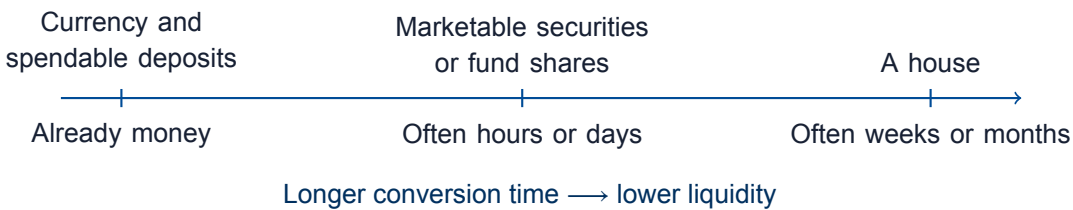
Money and Liquidity

- **Money**: what people readily accept as payment for goods and services.



Money and Liquidity

- **Money**: what people readily accept as payment for goods and services.
- **Liquidity**: how quickly something can become spendable money, with little cost or loss of value.



The Money Supply: M1 and M2

- **Money supply**: the total amount of currency and bank balances counted as money.

M2 = M1 + additional near-money. Near-money is readily convertible into spendable balances.

Source: Federal Reserve Board, H.6 definitions.

The Money Supply: M1 and M2

- **Money supply**: the total amount of currency and bank balances counted as money.
- **M1, the narrower measure**: public currency, demand deposits, and other liquid deposits, including savings deposits.

M2 = M1 + additional near-money. Near-money is readily convertible into spendable balances.

Source: Federal Reserve Board, H.6 definitions.

The Money Supply: M1 and M2

- **Money supply**: the total amount of currency and bank balances counted as money.
- **M1, the narrower measure**: public currency, demand deposits, and other liquid deposits, including savings deposits.
- **M2, the broader measure**: M1 plus small time deposits and retail money market mutual fund balances.

M2 = M1 + additional near-money. Near-money is readily convertible into spendable balances.

Source: Federal Reserve Board, H.6 definitions.

The Functions of Money

- **Medium of exchange**: an accepted way to pay for goods and services.
  - Pay for groceries without a double coincidence of wants.

The Functions of Money

- **Medium of exchange**: an accepted way to pay for goods and services.
  - Pay for groceries without a double coincidence of wants.
- **Unit of account**: a common unit for stating and comparing values.
  - Compare a $5 smoothie with a $10 sandwich.

The Functions of Money

- **Medium of exchange**: an accepted way to pay for goods and services.
  - Pay for groceries without a double coincidence of wants.
- **Unit of account**: a common unit for stating and comparing values.
  - Compare a $5 smoothie with a $10 sandwich.
- **Store of value**: a way to carry purchasing power into the future.
  - Save today's earnings for next month's groceries. Inflation can erode their value.

Types of Money

A farmer brings produce to market. Compare three ways to pay.



Producing crops: Bruegel, *The Harvesters*, 1565



Trading: Van Ruysdael, *Market by the Seashore*, 1637

Commodity money

Pay with a portion of the crop accepted as money. It also has value as food.

Images: The Met, 19.164 and The Met, 60.55.4; both CC0.

Types of Money

A farmer brings produce to market. Compare three ways to pay.



Producing crops: Bruegel, *The Harvesters*, 1565



Trading: Van Ruysdael, *Market by the Seashore*, 1637

Commodity money
Pay with a portion of the crop accepted as money. It also has value as food.

Commodity-backed money
Pay with a warehouse receipt redeemable for a specified quantity of grain.

Images: The Met, 19.164 and The Met, 60.55.4; both CC0.

Types of Money

A farmer brings produce to market. Compare three ways to pay.



Producing crops: Bruegel, *The Harvesters*, 1565



Trading: Van Ruysdael, *Market by the Seashore*, 1637

Commodity money
Pay with a portion of the crop accepted as money. It also has value as food.

Commodity-backed money
Pay with a warehouse receipt redeemable for a specified quantity of grain.

Fiat money
Pay with officially accepted money that promises no commodity redemption.

Images: The Met, 19.164 and The Met, 60.55.4; both CC0.

What Banks Do

- Banks **accept deposits** and provide payment services.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

What Banks Do

- Banks **accept deposits** and provide payment services.
- They keep part of their funding in **reserves** and lend a large portion or buy investments such as bonds.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

What Banks Do

- Banks **accept deposits** and provide payment services.
- They keep part of their funding in **reserves** and lend a large portion or buy investments such as bonds.
- Deposits can be withdrawn quickly; many loans and investments repay over years.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

Bank Reserves

- **Bank reserves**: cash in bank vaults plus the bank's deposits at the Federal Reserve.

Source: Federal Reserve Board, The Fed Explained.
Source: Federal Reserve Board, Reserve Requirements.

Bank Reserves

- **Bank reserves**: cash in bank vaults plus the bank's deposits at the Federal Reserve.
- Reserves let a bank meet withdrawals and settle payments to other banks.

Source: Federal Reserve Board, The Fed Explained.
Source: Federal Reserve Board, Reserve Requirements.

Bank Reserves

- **Bank reserves**: cash in bank vaults plus the bank's deposits at the Federal Reserve.
- Reserves let a bank meet withdrawals and settle payments to other banks.
- **Reserve ratio**: bank reserves divided by the bank's deposits.

Source: Federal Reserve Board, The Fed Explained.
Source: Federal Reserve Board, Reserve Requirements.

T-Accounts

A **T-account** shows **what a bank has on the left** and **what it owes on the right**.

What the bank has		What the bank owes	
Loans	$1,200,000	Deposits	$1,000,000
Reserves	$100,000		

Total on the left: $1,300,000. Deposits: $1,000,000. The $300,000 difference is owners' money, which absorbs losses first.

Numerical example: Krugman and Wells, Macroeconomics, 7th ed., Chapter 14.

T-Accounts

A **T-account** shows **what a bank has on the left** and **what it owes on the right**.

What the bank has		What the bank owes	
Loans	$1,200,000	Deposits	$1,000,000
Reserves	$100,000		

Total on the left: $1,300,000. Deposits: $1,000,000. The $300,000 difference is owners' money, which absorbs losses first.

Reserve ratio $= \frac{100{,}000}{1{,}000{,}000} = 10\%.$

Numerical example: Krugman and Wells, Macroeconomics, 7th ed., Chapter 14.

Money and Banking Checkpoint

A customer deposits $100 of currency into a checking account.

What changes for the bank and for public currency + deposits?

Money and Banking Checkpoint

A customer deposits $100 of currency into a checking account.

What changes for the bank and for public currency + deposits?

Reserves +$100; deposits +$100.

Money and Banking Checkpoint

A customer deposits $100 of currency into a checking account.

What changes for the bank and for public currency + deposits?

Reserves +$100; deposits +$100.

Public currency −$100, so currency + deposits is unchanged.

Bank Runs

- A **bank run** happens when many customers rush to take out their money because they fear the bank may fail.

Source: Federal Reserve Board, SVB review, April 2023.

Bank Runs

- A **bank run** happens when many customers rush to take out their money because they fear the bank may fail.
- The bank has lent or invested much of that money. It cannot get it all back immediately.

Source: Federal Reserve Board, SVB review, April 2023.

Bank Runs

- A **bank run** happens when many customers rush to take out their money because they fear the bank may fail.
- The bank has lent or invested much of that money. It cannot get it all back immediately.
- Selling investments quickly can mean losses. This can make more customers want to leave.

Source: Federal Reserve Board, SVB review, April 2023.

Silicon Valley Bank: March 2023

- **2022**: higher interest rates made SVB's older, long-term bonds worth less.

"Silicon Valley Bank shut by California regulator"
Reuters, March 10, 2023.

Source: Federal Reserve Board, SVB review, April 2023.

Silicon Valley Bank: March 2023

- **2022**: higher interest rates made SVB's older, long-term bonds worth less.
- **March 8–9**: SVB announced losses. Worried customers withdrew more than $40 billion the next day.

"Silicon Valley Bank shut by California regulator"
Reuters, March 10, 2023.

Source: Federal Reserve Board, SVB review, April 2023.

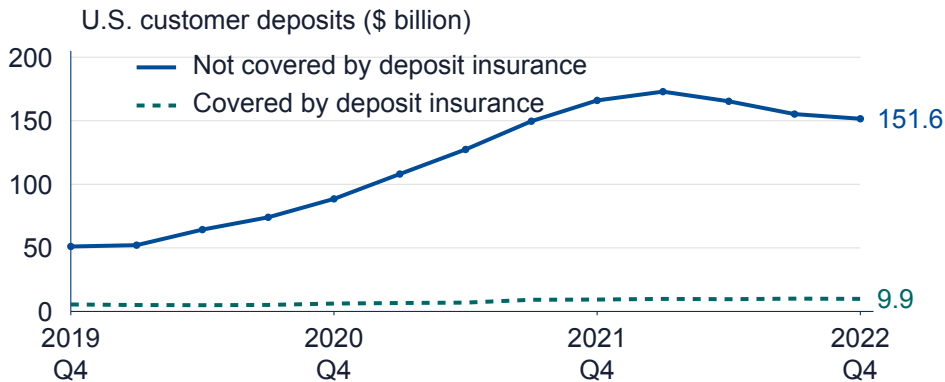
Silicon Valley Bank: March 2023

- **2022**: higher interest rates made SVB's older, long-term bonds worth less.
- **March 8–9**: SVB announced losses. Worried customers withdrew more than $40 billion the next day.
- **March 10**: the bank could not meet the rush of withdrawals. Regulators closed it.

"Silicon Valley Bank shut by California regulator"
Reuters, March 10, 2023.

Source: Federal Reserve Board, SVB review, April 2023.

Silicon Valley Bank: Customer Deposits



Original graph. Data: Federal Reserve Board, April 2023 SVB review, Figure 7 data; Call Report.

Bank Regulation

- **Deposit insurance**: the FDIC, created in 1933, protects eligible deposits up to $250,000 per depositor, insured bank, and ownership category.

Sources: FDIC history; deposit coverage; Federal Reserve: discount window, IORB, capital.

Bank Regulation

- **Deposit insurance**: the FDIC, created in 1933, protects eligible deposits up to $250,000 per depositor, insured bank, and ownership category.
- **Discount window**: the Fed lends a bank money to pay withdrawals. The loan must be backed by acceptable loans or bonds and repaid with interest.

Sources: FDIC history; deposit coverage; Federal Reserve: discount window, IORB, capital.

Bank Regulation

- **Deposit insurance**: the FDIC, created in 1933, protects eligible deposits up to $250,000 per depositor, insured bank, and ownership category.
- **Discount window**: the Fed lends a bank money to pay withdrawals. The loan must be backed by acceptable loans or bonds and repaid with interest.
- **Interest on reserve balances (IORB)**: the Fed pays interest on money banks keep in their accounts at the Fed.

Sources: FDIC history; deposit coverage; Federal Reserve: discount window, IORB, capital.

Bank Regulation

- **Deposit insurance**: the FDIC, created in 1933, protects eligible deposits up to $250,000 per depositor, insured bank, and ownership category.
- **Discount window**: the Fed lends a bank money to pay withdrawals. The loan must be backed by acceptable loans or bonds and repaid with interest.
- **Interest on reserve balances (IORB)**: the Fed pays interest on money banks keep in their accounts at the Fed.
- **Capital requirements**: owners must put in their own money to absorb losses first.

Sources: FDIC history; deposit coverage; Federal Reserve: discount window, IORB, capital.

Moral Hazard

- **Moral hazard**: protection from losses can encourage greater risk-taking.

Source: Federal Reserve Board, regulatory capital.

Moral Hazard

- **Moral hazard**: protection from losses can encourage greater risk-taking.
- Customers may watch risks less closely if their deposits are insured. Owners may gain from risky bets while others bear some losses.

Source: Federal Reserve Board, regulatory capital.

Moral Hazard

- **Moral hazard**: protection from losses can encourage greater risk-taking.
- Customers may watch risks less closely if their deposits are insured. Owners may gain from risky bets while others bear some losses.
- **Capital requirements** make owners risk more of their own money; bank oversight helps enforce the rules.

Source: Federal Reserve Board, regulatory capital.

Shadow Banking

- **Shadow banking**: lending and borrowing through firms that are not ordinary deposit banks.

Source: Financial Stability Board, Non-Bank Financial Intermediation.

Source: Federal Reserve Board, crisis institutions and funding runs (2009).

Shadow Banking

- **Shadow banking**: lending and borrowing through firms that are not ordinary deposit banks.
- Examples: **investment banks** help companies raise money; **money market funds** pool savings for short-term lending.

Source: Financial Stability Board, Non-Bank Financial Intermediation.

Source: Federal Reserve Board, crisis institutions and funding runs (2009).

Shadow Banking

- **Shadow banking**: lending and borrowing through firms that are not ordinary deposit banks.
- Examples: **investment banks** help companies raise money; **money market funds** pool savings for short-term lending.

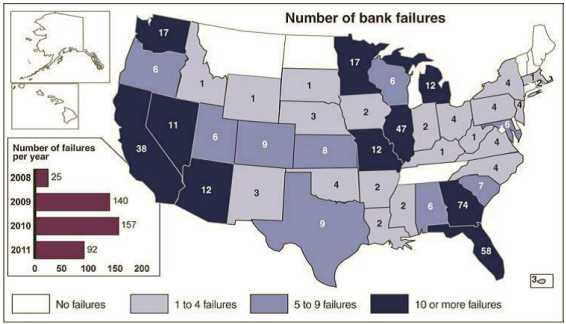
The 2008 crisis

- September 15: **Lehman Brothers** went bankrupt.
- September 16: losses at **Reserve Primary Fund** led investors to demand their money back.

Source: Financial Stability Board, Non-Bank Financial Intermediation.

Source: Federal Reserve Board, crisis institutions and funding runs (2009).

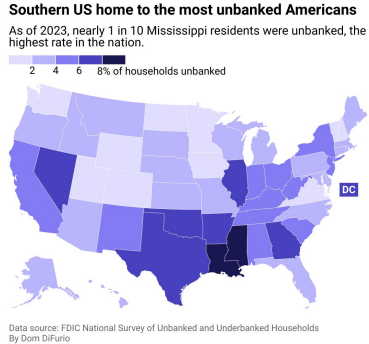
Bank Failures and Unbanked Households



Source: GAO analysis of FDIC data: Map Resources (map).

Map: GAO analysis of FDIC data, 2008–2011.

Reference: FDIC (2020), Staff Studies 2020–05, Table 1.



FDIC (2024), 2023 Household Survey, state estimates.

Visual: Dom DiFurio. Measure: share of households.

Reference Definitions and Examples

Money, income, wealth, and bank accounting.

Money, Income, and Net Wealth

Concept	Meaning	Example
Money	Readily spendable assets	$400 in currency and deposits
Income	Receipts over a period	$900 earned this month
Net wealth	Assets minus liabilities	$5,000 − $2,000 = $3,000

A Smoothie Company Balance Sheet

Assets	Liabilities and equity
Cash and inventory	Loans and other obligations
Equipment; a building if owned	Equity: the owners' residual claim

Assets = liabilities + equity.

Liquidity and Solvency

Issue	Question	Possible problem
Liquidity	Can the bank pay now?	Assets cannot all be sold quickly at little cost.
Solvency	Do assets cover liabilities?	Losses can exhaust equity.

Forced sales can turn liquidity pressure into losses.

A Cash Deposit

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

A customer deposits $200 of currency held outside banks.

A Cash Deposit

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

A customer deposits $200 of currency held outside banks.

Reserves and deposits each rise by $200.

A Cash Deposit

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

A customer deposits $200 of currency held outside banks.

Reserves and deposits each rise by $200.

Public currency falls by $200; currency + deposits is unchanged.

A New Bank Loan

- The bank approves a $120 loan and credits the borrower's deposit.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

A New Bank Loan

- The bank approves a $120 loan and credits the borrower's deposit.
- Loans and deposits each rise by $120 at origination.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

A New Bank Loan

- The bank approves a $120 loan and credits the borrower's deposit.
- Loans and deposits each rise by $120 at origination.
- The borrower gains a deposit and owes a debt; net wealth does not rise by $120.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

Reserves and the Monetary Base

- **Public currency**: notes and coins outside banks.

Source: Federal Reserve Board, H.6 definitions.

Reserves and the Monetary Base

- **Public currency**: notes and coins outside banks.
- **Bank reserves**: vault cash and balances at the Fed.

Source: Federal Reserve Board, H.6 definitions.

Reserves and the Monetary Base

- **Public currency**: notes and coins outside banks.
- **Bank reserves**: vault cash and balances at the Fed.
- **Monetary base**: public currency plus bank reserves.

Source: Federal Reserve Board, H.6 definitions.